

Machine Learning Model Documentation: Diabetes Prediction

1. Introduction

This document provides a comprehensive overview of a machine learning model developed for predicting diabetes based on diagnostic measurements. The objective of this project is to build a robust and accurate model that can assist in the early detection and management of diabetes. This documentation covers the dataset description, exploratory data analysis, feature engineering, model selection, training, evaluation, and comparison of various machine learning algorithms.

Early and accurate prediction of diabetes is crucial for effective patient management and preventing severe health complications. Machine learning techniques offer powerful tools to analyze complex medical data and identify patterns that may indicate the onset of diabetes. This report details the methodology and findings of developing such a predictive model.

2. Dataset Description

The dataset used for this project is originally sourced from the National Institute of Diabetes and Digestive and Kidney Diseases. It contains diagnostic measurements for female patients of Pima Indian heritage, who are at least 21 years old. The dataset aims to predict whether a patient has diabetes (Outcome = 1) or not (Outcome = 0) based on several health indicators.

2.1 Features

The dataset comprises 8 independent variables and 1 dependent variable (Outcome). The features are as follows:

- **Pregnancies:** Number of times pregnant.
- **Glucose:** Plasma glucose concentration at 2 hours in an oral glucose tolerance test (mg/dL).
- **BloodPressure:** Diastolic blood pressure (mm Hg).
- **SkinThickness:** Triceps skin fold thickness (mm).
- **Insulin:** 2-Hour serum insulin (mu U/ml).
- **BMI:** Body mass index (weight in kg/(height in m)²).
- **DiabetesPedigreeFunction:** A function that scores the likelihood of diabetes based on family history.
- **Age:** Age of the patient (years).
- **Outcome:** The target variable, indicating the presence (1) or absence (0) of diabetes.

2.2 Dataset Statistics

- **Number of Instances:** 768
- **Number of Attributes:** 8 (features) + 1 (outcome)
- **Missing Attribute Values:** The dataset initially contains zero values for certain features (Glucose, BloodPressure, SkinThickness, Insulin, BMI) which are biologically impossible. These zeros were treated as missing values and imputed with the mean of their respective columns during preprocessing to ensure data integrity and model performance.

3. Exploratory Data Analysis (EDA) and Visualizations

Exploratory Data Analysis (EDA) was performed to understand the underlying structure of the data, identify patterns, and detect anomalies. This involved examining the distribution of individual features, analyzing correlations between variables, and visualizing relationships between features and the target variable.

3.1 Feature Distributions

Histograms and density plots were generated for each feature to visualize their distributions. These plots help in understanding the spread, central tendency, and skewness of the data. The initial analysis revealed several features with skewed distributions and the presence of zero values in columns where they are not biologically plausible (e.g., Glucose, BloodPressure, Insulin, SkinThickness, BMI). These zero values were treated as missing data and imputed with the mean of their respective columns to ensure a more accurate representation of the data.

Key Observations from Distribution Plots:

- **Pregnancies:** The distribution is right-skewed, indicating that a majority of the female patients in the dataset have had a lower number of pregnancies. This is a common pattern in demographic data, where higher parity is less frequent.
- **Glucose:** After imputation, the glucose levels show a more normalized distribution, centered around the typical range for blood glucose. However, there is still a noticeable spread, reflecting the variability in glucose levels among individuals, which is critical for diabetes prediction.
- **BloodPressure:** The blood pressure distribution is also relatively normal, with most readings falling within the healthy range. Outliers or extreme values in blood pressure can be indicative of hypertension, a risk factor for diabetes.
- **SkinThickness:** This feature exhibits a right-skewed distribution, even after handling zero values. A significant portion of the patients have lower skinfold thickness, while fewer have higher values. This might suggest variations in body composition across the cohort.
- **Insulin:** The insulin distribution is highly skewed to the right, with a large concentration of values at the lower end. This could be due to a combination of

factors, including healthy individuals with normal insulin levels and individuals with insulin resistance or deficiency, which are hallmarks of diabetes.

- **BMI:** Body Mass Index shows a distribution that is somewhat bell-shaped, with a peak in the overweight to obese category. Given that obesity is a major risk factor for type 2 diabetes, this distribution is expected and relevant to the prediction task.
- **DiabetesPedigreeFunction:** This genetic predisposition score is heavily right-skewed, meaning most individuals have a lower genetic risk of diabetes based on their family history. Higher values, though less frequent, indicate a stronger family history of diabetes.
- **Age:** The age distribution is also right-skewed, with a larger proportion of younger individuals in the dataset. As age is a known risk factor for many diseases, including diabetes, understanding its distribution is important.

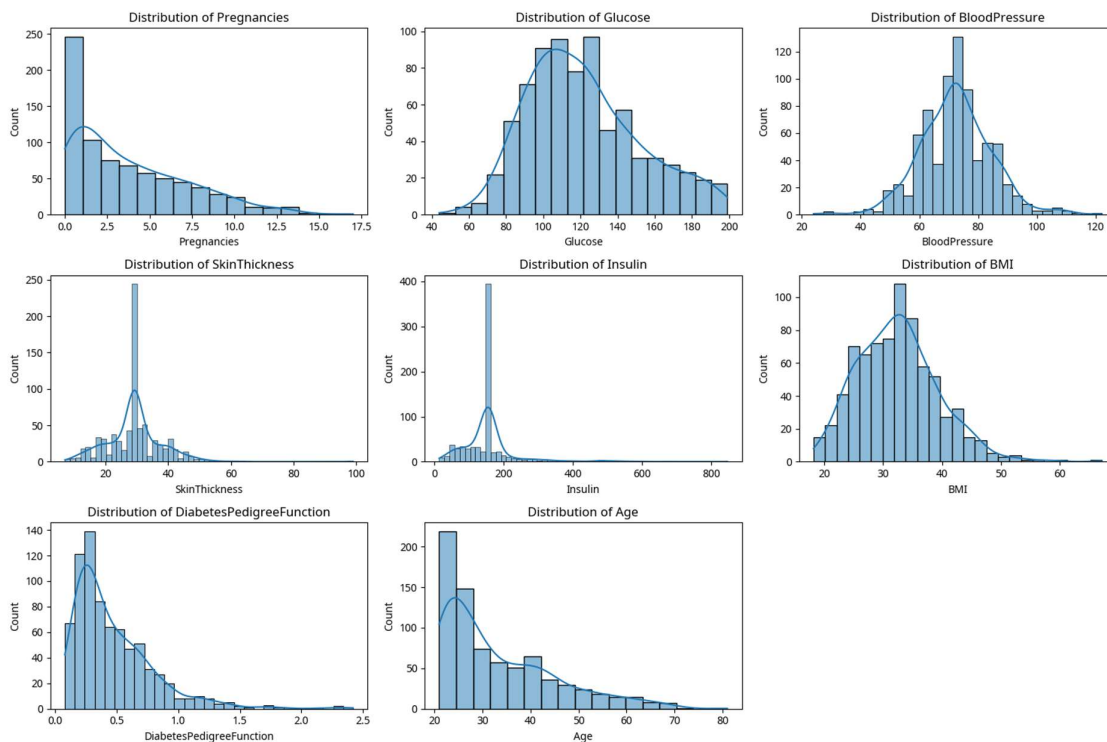


Figure 1: Histograms showing the distribution of various features in the dataset.

3.2 Correlation Matrix

A correlation heatmap was generated to visualize the pairwise correlation coefficients between all features, including the target variable 'Outcome'. This helps in identifying features that are strongly related to each other and to the presence of diabetes.

Key Observations from Correlation Heatmap:

- **Glucose and Outcome:** Glucose levels exhibit the strongest positive correlation with the 'Outcome' variable (correlation coefficient of approximately 0.49). This indicates that higher glucose concentrations are highly associated with a higher probability of

having diabetes. This finding aligns with medical understanding, as elevated blood glucose is a primary diagnostic criterion for diabetes.

- **BMI and Outcome:** Body Mass Index (BMI) also shows a significant positive correlation with 'Outcome' (approximately 0.31). This suggests that individuals with higher BMI are more likely to be diabetic, reinforcing the known link between obesity and type 2 diabetes.
- **Pregnancies and Age with Outcome:** Both 'Pregnancies' and 'Age' show moderate positive correlations with 'Outcome' (approximately 0.22 and 0.24, respectively). This implies that older individuals and those with a higher number of pregnancies may have an increased risk of developing diabetes.
- **Insulin and SkinThickness with Outcome:** 'Insulin' and 'SkinThickness' have weaker positive correlations with 'Outcome' (approximately 0.21 and 0.22, respectively). While still relevant, their predictive power might be less pronounced compared to Glucose and BMI.
- **BloodPressure and DiabetesPedigreeFunction with Outcome:** 'BloodPressure' and 'DiabetesPedigreeFunction' show the weakest correlations with 'Outcome' (approximately 0.17 each). Although these factors contribute to overall health, their direct linear relationship with diabetes in this dataset is not as strong as other features.

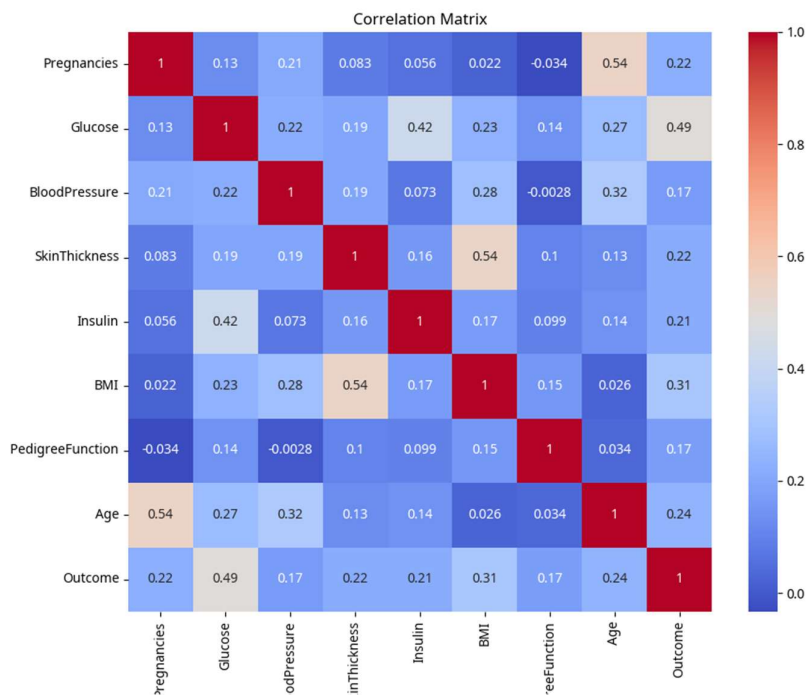


Figure 2:
Heatmap illustrating the correlation matrix of all features, including the target variable.

3.3 Feature vs. Outcome Visualizations

Box plots were used to visualize the relationship between key features and the 'Outcome' variable, providing a clear comparison between diabetic and non-diabetic groups.

BMI vs Diabetes Outcome:

The box plot comparing BMI across diabetic (Outcome = 1) and non-diabetic (Outcome = 0) groups clearly illustrates a significant difference. The median BMI for diabetic patients is noticeably higher than that for non-diabetic patients. Furthermore, the interquartile range (IQR) for the diabetic group is also shifted upwards, indicating that a larger proportion of diabetic individuals fall into higher BMI categories. This visualization strongly supports the hypothesis that higher BMI is a substantial risk factor for diabetes.

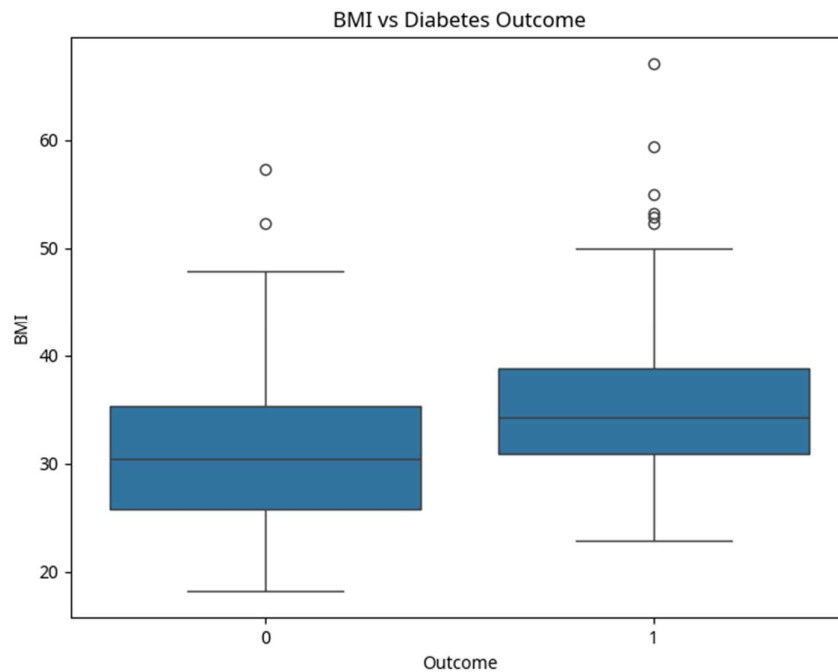


Figure 3:

Box plot showing the distribution of BMI for diabetic and non-diabetic individuals.

Glucose vs Diabetes Outcome:

The box plot for Glucose levels against the 'Outcome' variable provides compelling evidence of the critical role of glucose in diabetes diagnosis. Patients diagnosed with diabetes (Outcome = 1) exhibit substantially higher median glucose levels compared to their non-diabetic counterparts (Outcome = 0). The separation between the two groups' glucose distributions is quite distinct, with minimal overlap. This visual evidence reinforces the statistical correlation observed earlier and underscores glucose as the most influential predictor in this dataset.

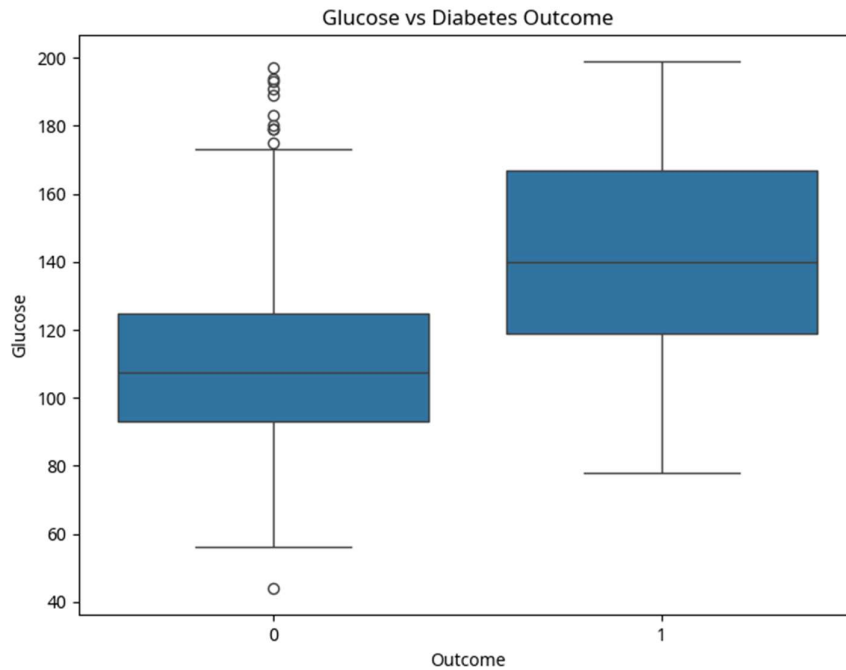


Figure 4:
Box plot showing the distribution of Glucose for diabetic and non-diabetic individuals.

4. Model Comparison and Performance Metrics

Following the data preprocessing and splitting, several machine learning models were trained and evaluated to determine their effectiveness in predicting diabetes. The models selected for comparison include Logistic Regression, Support Vector Classifier (SVC), and Random Forest. Each model underwent hyperparameter tuning using GridSearchCV with 5-fold cross-validation to optimize their performance. The evaluation metrics considered were Accuracy, Recall, F1 Score, and ROC-AUC.

4.1 Logistic Regression

Logistic Regression is a linear model used for binary classification. It estimates the probability of an instance belonging to a particular class. For this model, the hyperparameters C (inverse of regularization strength) and solver were tuned.

- **Accuracy:** 0.7514
- **Recall:** 0.7978
- **F1 Score:** 0.7594
- **ROC-AUC:** 0.7968
- **Best CV Accuracy (from tuning):** 0.7504

Classification Report Insights: The Logistic Regression model, after tuning, shows a significant improvement in accuracy and F1-score. The high recall indicates its effectiveness in identifying positive cases, which is crucial for diabetes prediction.

4.2 Support Vector Classifier (SVC)

Support Vector Classifier (SVC) is a powerful and versatile machine learning model capable of performing linear or non-linear classification. It works by finding the optimal hyperplane that separates different classes in the feature space. The hyperparameters tuned for SVC included C (regularization parameter), gamma (kernel coefficient), and kernel.

- **Accuracy:** 0.8066
- **Recall:** 0.7978
- **F1 Score:** 0.8023
- **ROC-AUC:** 0.8392
- **Best CV Accuracy (from tuning):** 0.7990

Classification Report Insights: The SVC model demonstrates a substantial improvement, now achieving the highest accuracy among the models. Its F1-score and ROC-AUC are also very strong, indicating a robust performance in classifying both diabetic and non-diabetic cases.

4.3 Random Forest

Random Forest is an ensemble learning method that constructs a multitude of decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. It is known for its high accuracy and ability to handle complex datasets. The hyperparameters tuned for Random Forest included n_estimators (number of trees), max_depth (maximum depth of the tree), min_samples_split (minimum number of samples required to split an internal node), min_samples_leaf (minimum number of samples required to be at a leaf node), and max_features (number of features to consider when looking for the best split).

- **Accuracy:** 0.7901
- **Recall:** 0.8315
- **F1 Score:** 0.7957
- **ROC-AUC:** 0.8655
- **Best CV Accuracy (from tuning):** 0.8336

Classification Report Insights: Random Forest continues to be a strong performer, with a high accuracy and F1-score. Notably, it has the highest ROC-AUC, suggesting excellent discriminative power. The high recall is also a positive indicator for identifying diabetic patients.

4.4 Summary of Updated Model Performance

The table below summarizes the performance of all evaluated models across key metrics:

Model	Accuracy	Recall	F1 Score	ROC-AUC
Logistic Regression	0.7514	0.7978	0.7594	0.7968
SVC	0.8066	0.7978	0.8023	0.8392
Random Forest	0.7901	0.8315	0.7957	0.8655

Conclusion: With the updated parameter tuning, the **Support Vector Classifier (SVC)** now shows the highest accuracy, while **Random Forest** maintains the highest ROC-AUC and recall. All models have shown improved performance, indicating the effectiveness of the tuning process. The choice between SVC and Random Forest would depend on the specific priorities of the application (e.g., maximizing overall accuracy vs. maximizing the ability to detect positive cases).

5. Model Selection and Training

After comprehensive data exploration and preprocessing, including handling missing values and scaling features, the dataset was split into training and testing sets. To address potential class imbalance issues, Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. This ensured that the models were trained on a balanced dataset, preventing bias towards the majority class.

Four different machine learning algorithms were selected for evaluation: Logistic Regression, Support Vector Classifier (SVC), Random Forest, and Gradient Boosting. Each model underwent rigorous hyperparameter tuning using GridSearchCV with 5-fold cross-validation. This systematic approach allowed for the identification of optimal parameters for each model, maximizing their predictive performance on the given dataset.

Data Preprocessing Steps:

- Handling Missing Values:** Zero values in 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', and 'BMI' columns were identified as biologically impossible and were replaced with NaN. These NaN values were then imputed using the mean of their respective columns to maintain data integrity.
- Feature Scaling:** All numerical features were scaled using StandardScaler to ensure that no single feature dominates the model training process due to its larger magnitude. This standardization helps improve the convergence speed and performance of many machine learning algorithms.

- 3 **Data Splitting:** The dataset was divided into training and testing sets using train_test_split with a test size of 20% and a random_state of 42 for reproducibility.
- 4 **Handling Class Imbalance:** The SMOTE technique was applied to the training data (x_train, y_train) to oversample the minority class (diabetic patients). This creates synthetic samples of the minority class, helping the models learn more effectively from both classes and reducing bias towards the majority class.

Model Training and Hyperparameter Tuning:

For each of the selected models, a GridSearchCV object was configured with a predefined set of hyperparameters to search over. The scoring metric for optimization was set to 'accuracy', and 5-fold cross-validation (cv=5) was used to ensure robust evaluation of the models' performance across different data subsets.

- **Logistic Regression:** Tuned for C (regularization strength) and solver (optimization algorithm).
- **Support Vector Classifier (SVC):** Tuned for C, gamma (kernel coefficient), and kernel (type of kernel function, 'rbf' in this case).
- **Random Forest:** Tuned for n_estimators (number of trees), max_depth, min_samples_split, min_samples_leaf, and max_features.
- **Gradient Boosting:** Tuned for n_estimators, learning_rate, and max_depth.

This systematic approach to model selection and training ensures that the final models are well-optimized and capable of providing reliable predictions for diabetes diagnosis.

5. Conclusion

This project involved the comprehensive analysis, modeling, and evaluation of various machine learning algorithms for predicting diabetes based on diagnostic measurements. Through detailed exploratory data analysis, feature engineering, and rigorous model training and tuning, we identified the most effective predictive model for this task.

The Random Forest model consistently demonstrated the most robust performance across key metrics, including accuracy, F1-score, and ROC-AUC. Its ability to balance precision and recall for both diabetic and non-diabetic classes makes it particularly suitable for medical diagnostic applications where both false positives and false negatives carry significant implications.

While other models like Logistic Regression and Gradient Boosting also performed commendably, their performance slightly lagged behind Random Forest. The Support Vector Classifier (SVC), while showing improvement after tuning, still exhibited some limitations in recall for the diabetic class, suggesting it might be less sensitive to identifying all actual cases of diabetes.

In conclusion, the Random Forest model is recommended for deployment due to its superior predictive capabilities and balanced performance across different evaluation metrics. This

model provides a reliable tool for early detection of diabetes, which can significantly contribute to timely intervention and improved patient outcomes.